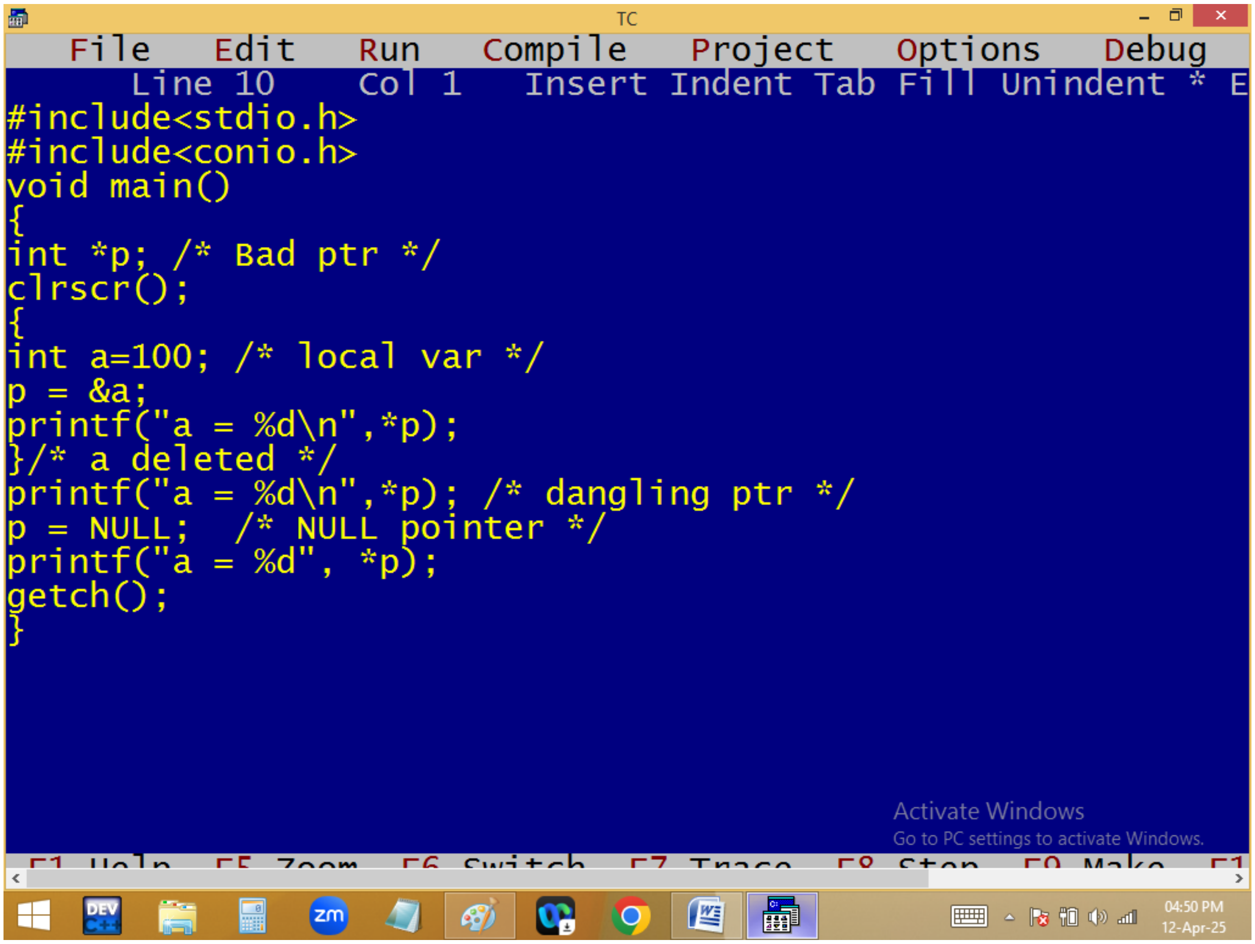
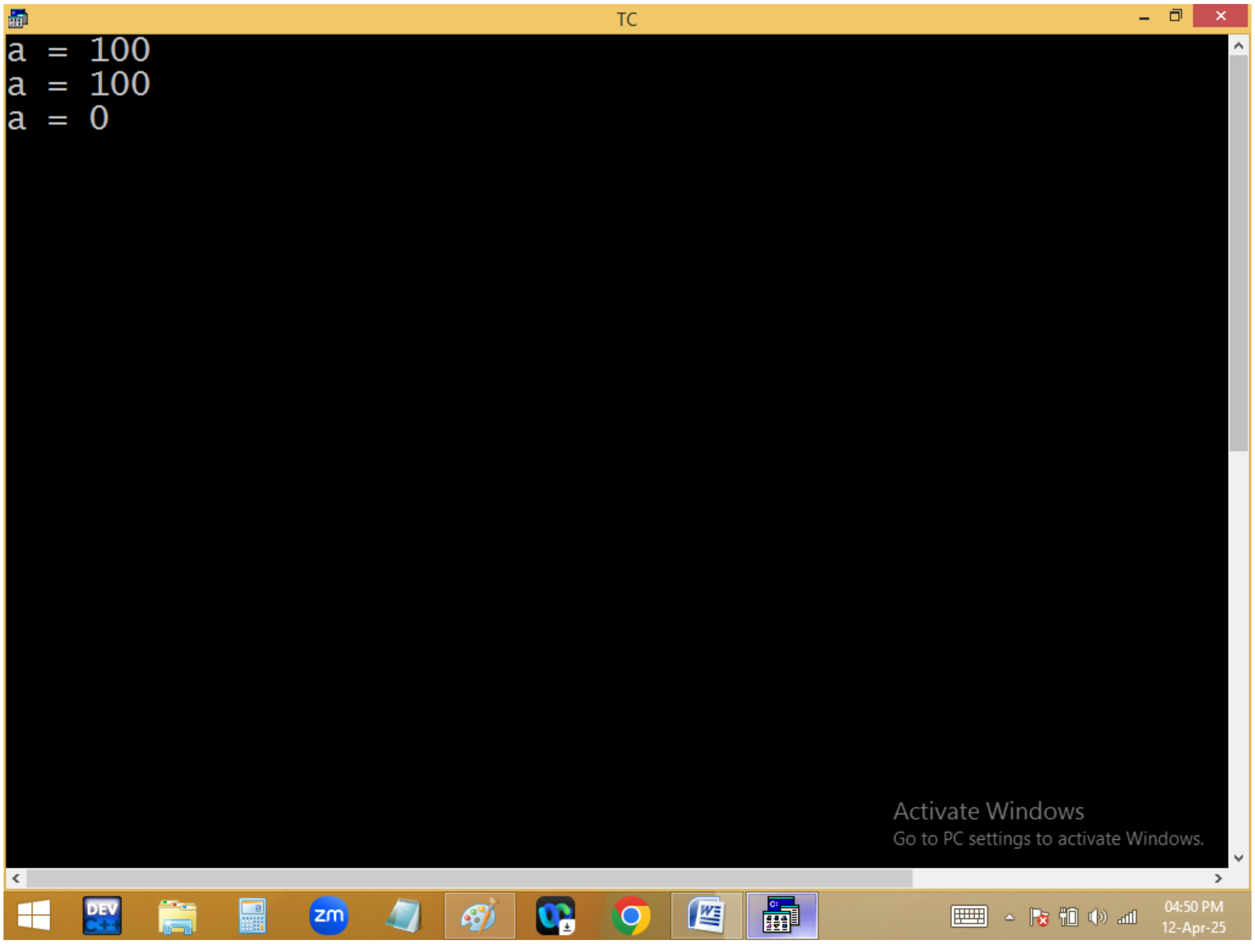
Dangling pointer:

A pointer is declared and some variable address also assigned. After some time that variable is deleted from memory. But the pointer is still pointing the deleted variable address. In this situation the pointer is called dangling pointer and to avoid this initialize the pointer with 0 or NULL.


TC
File Edit Run Compile Project Options Debug
Line 10 Col 1 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int *p; /* Bad ptr */
clrscr();
{
int a=100; /* local var */
p = &a;
printf("a = %d\n",*p);
}/* a deleted */
printf("a = %d\n",*p); /* dangling ptr */
p = NULL; /* NULL pointer */
printf("a = %d", *p);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
04:50 PM
12-Apr-25

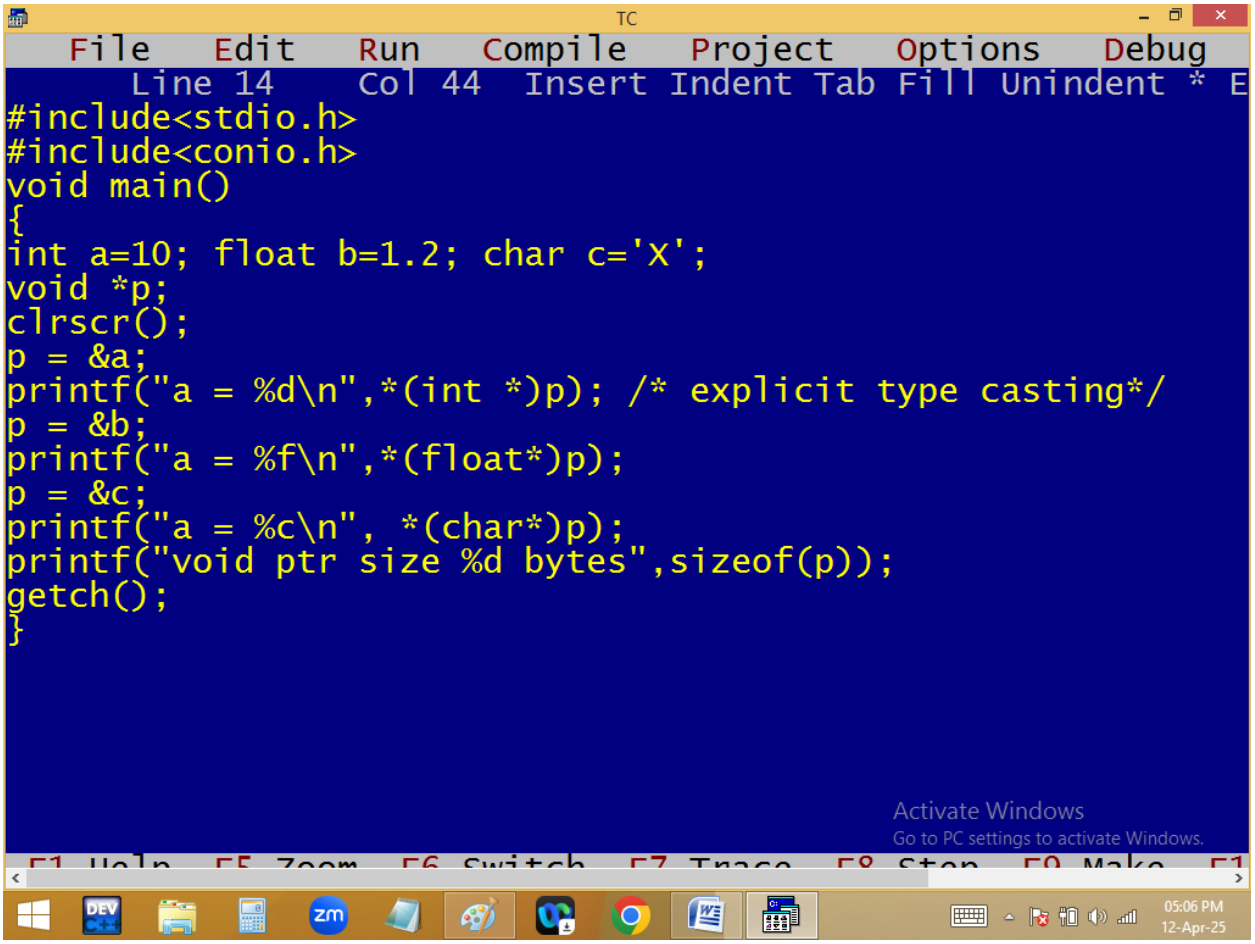


void / generic pointer: pointer can store only the same type of variable address. When several variables with different data type, we have to declare several pointers.

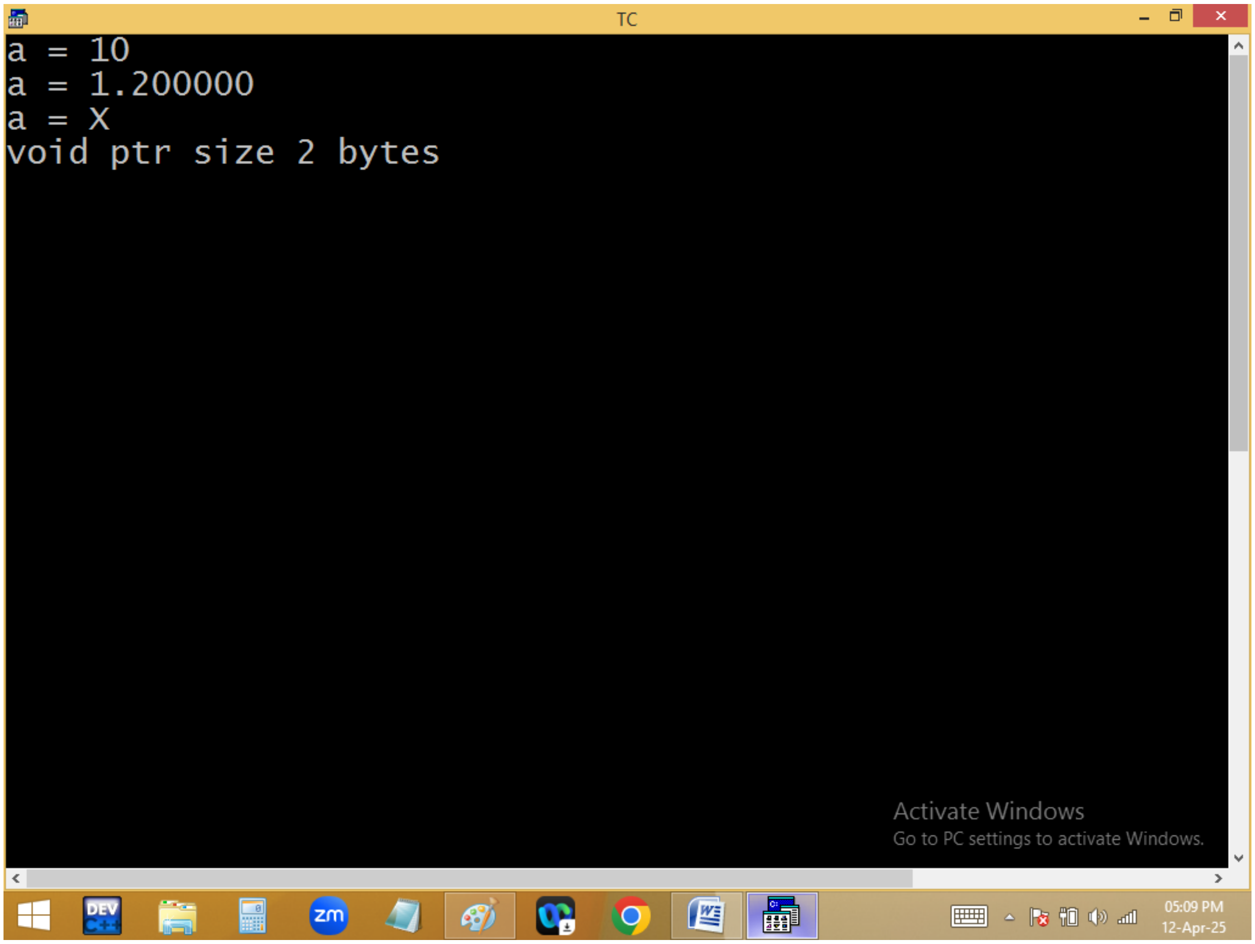
Void pointer can store any type of variable address. Before going access void pointer, explicit type casting should be provided.

void pointer takes 2 bytes.

Void pointer very much used in dynamic memory allocation.

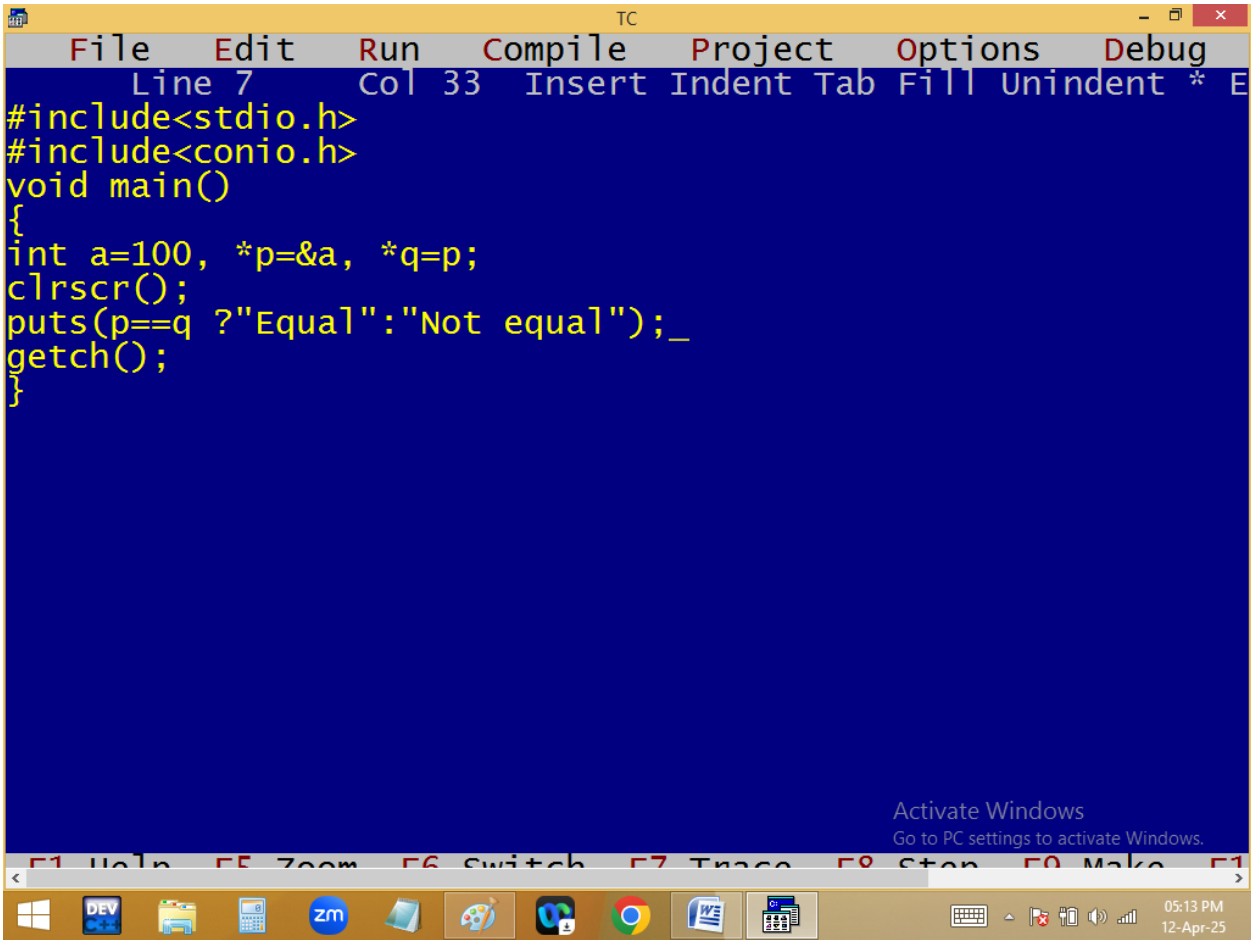


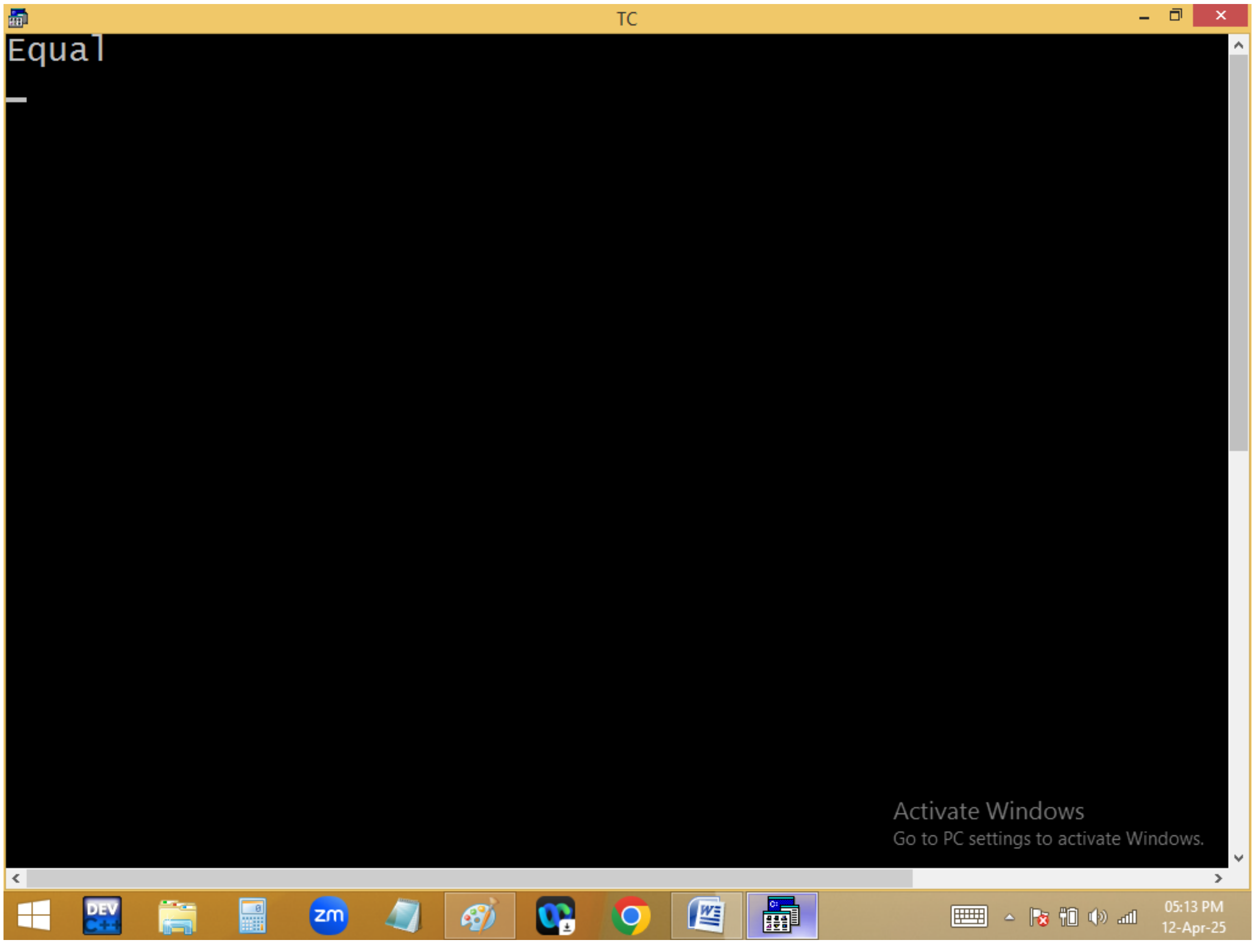

```c
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10; float b=1.2; char c='X';
void *p;
clrscr();
p = &a;
printf("a = %d\n",*(int *)p); /* explicit type casting*/
p = &b;
printf("a = %f\n",*(float*)p);
p = &c;
printf("a = %c\n", *(char*)p);
printf("void ptr size %d bytes",sizeof(p));
getch();
}
```

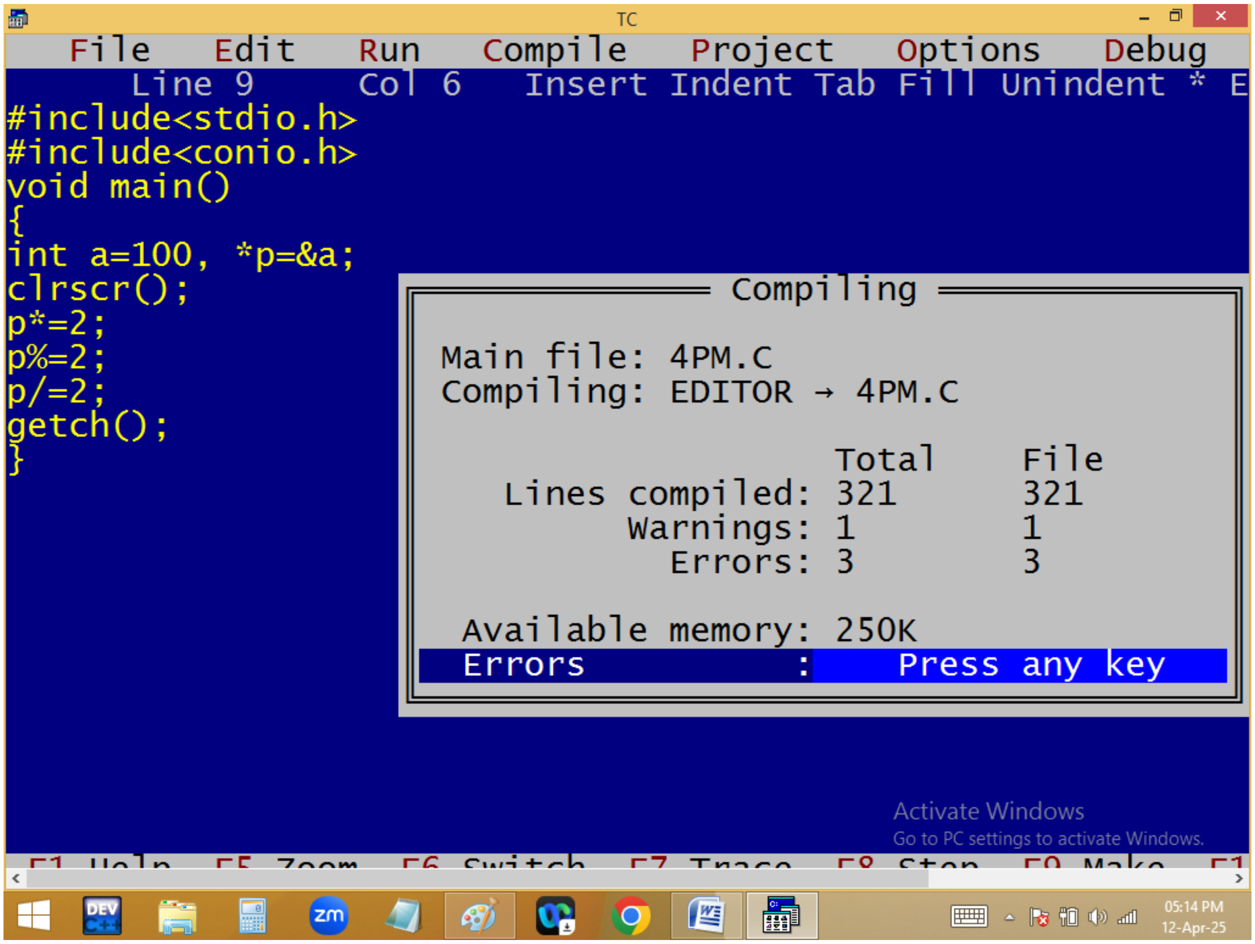


Pointer arithmetic:

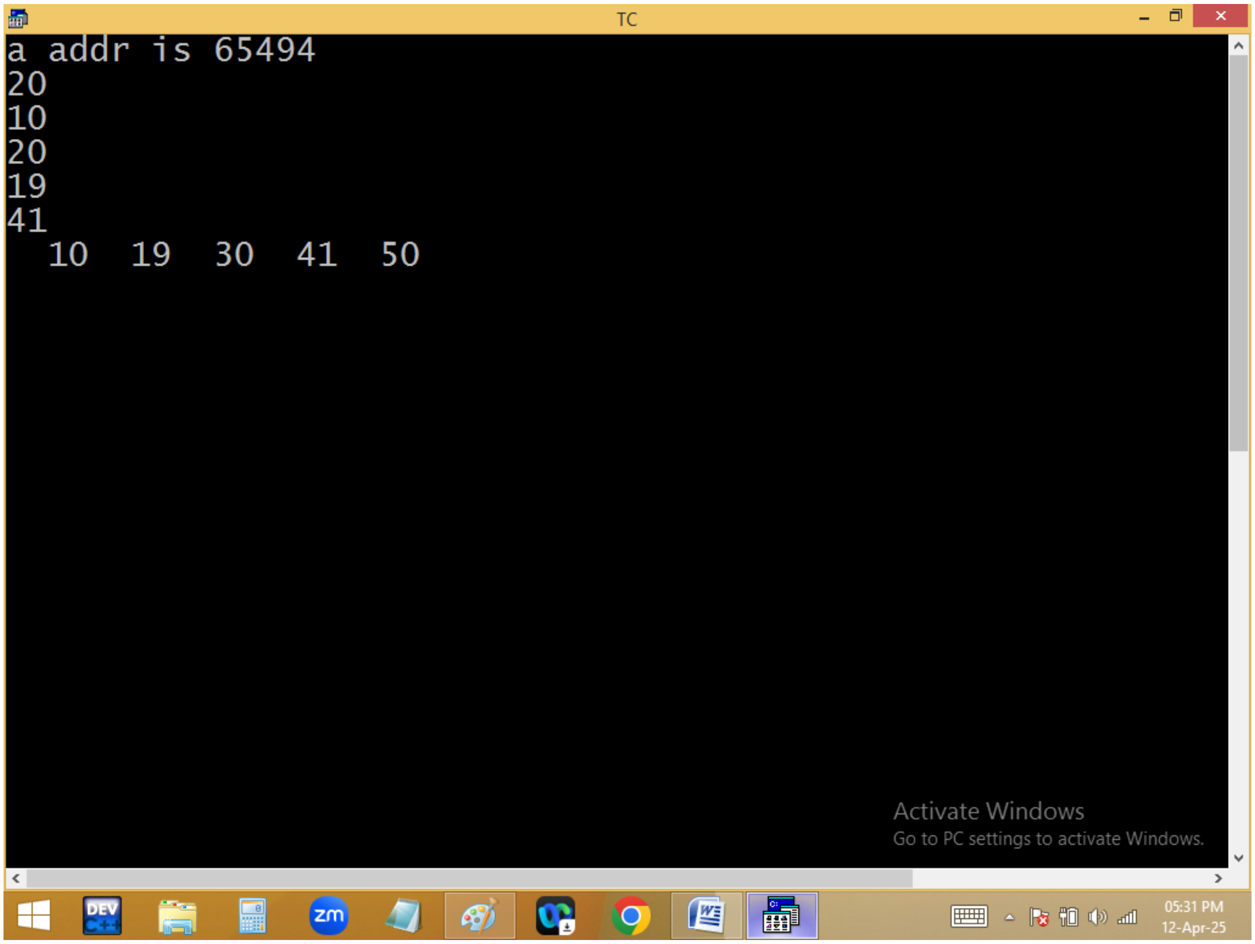
Like normal variables we can do +, -, ++, --, =, == on pointers also. But we can't do *, % , / on pointers.

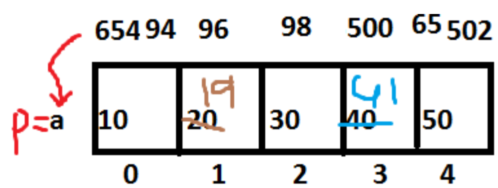

TC
File Edit Run Compile Project Options Debug
Line 7 Col 33 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=100, *p=&a, *q=p;
clrscr();
puts(p==q ?"Equal":"Not equal");_
getch();
}
Activate Windows
Go to PC settings to activate Windows.
05:13 PM
12-Apr-25


TC
Equal
Activate Windows
Go to PC settings to activate Windows.
05:13 PM
12-Apr-25


TC
File Edit Run Compile Project Options Debug
Line 9 Col 6 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=100, *p=&a;
clrscr();
p*=2;
p%=2;
p/=2;
getch();
}
Compiling
Main file: 4PM.C
Compiling: EDITOR → 4PM.C
Total File
Lines compiled: 321 321
Warnings: 1 1
Errors: 3 3
Available memory: 250K
Errors : Press any key
Activate Windows
Go to PC settings to activate Windows.
05:14 PM
12-Apr-25

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a[5]={10,20,30,40,50}, *p=&a[0],i;
clrscr();
printf("a addr is %u\n",a);
p++; printf("%d\n",*p);
--p; printf("%d\n",*p);
*p++; printf("%d\n",*p);
--*p; printf("%d\n",*p);
p+=2; (*p)++; printf("%d\n",*p);
for(i=0;i<5;i++)
printf("%4d",a[i]);
getch();
}
```





p++; 65494+1*2==>65496

***p ==> value at 65496 ==> 20**

--p; ==>65496-1*2==>65494

***p ==> value at 65494 ==> 10**

***p++==>p++==> 65494+1*2=65496**

***p ==> value at 65496==>20**

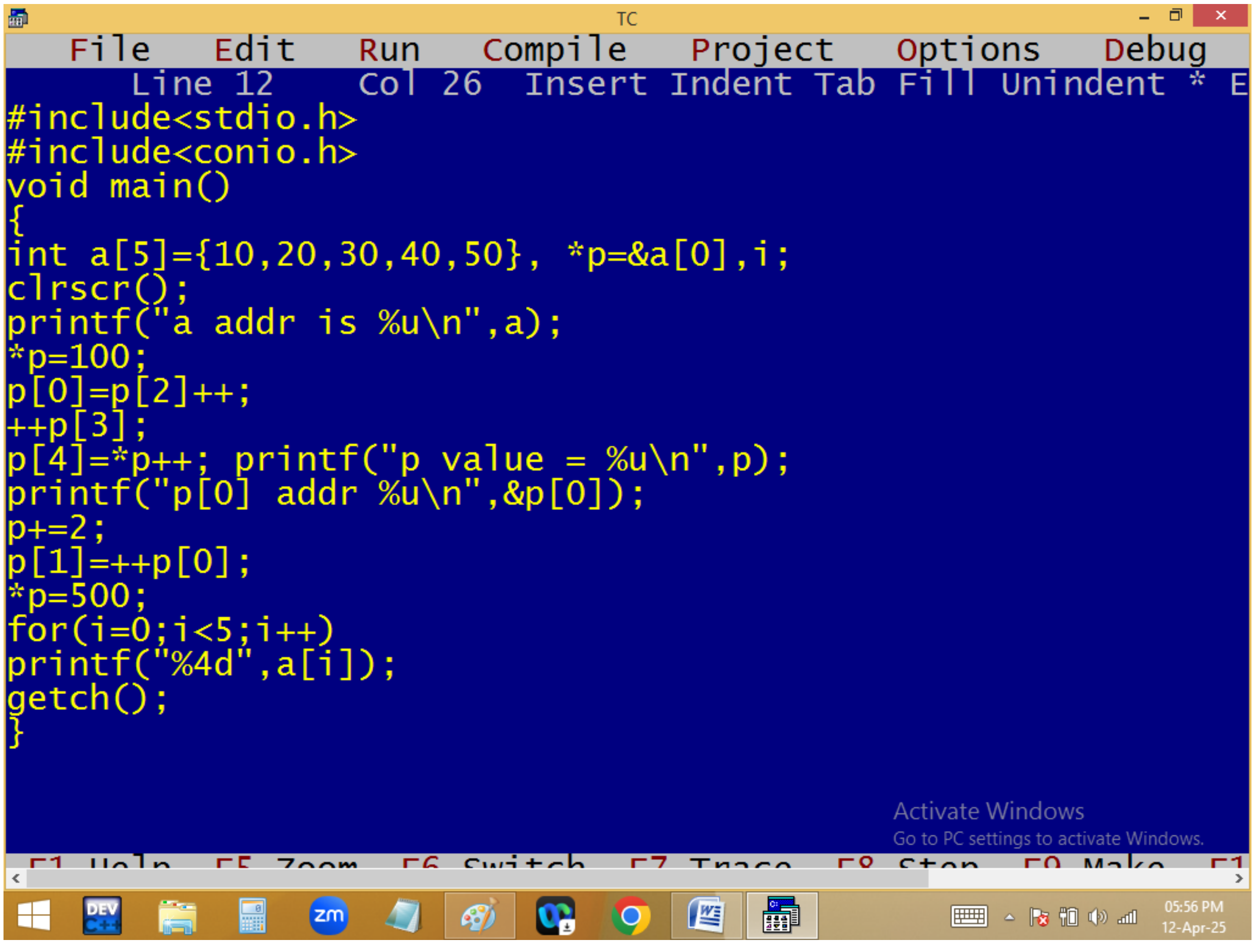
--*p ==> -- of value at 65496 ==>--20 ==> 19

***p ==>value at 65496 ==> 19**

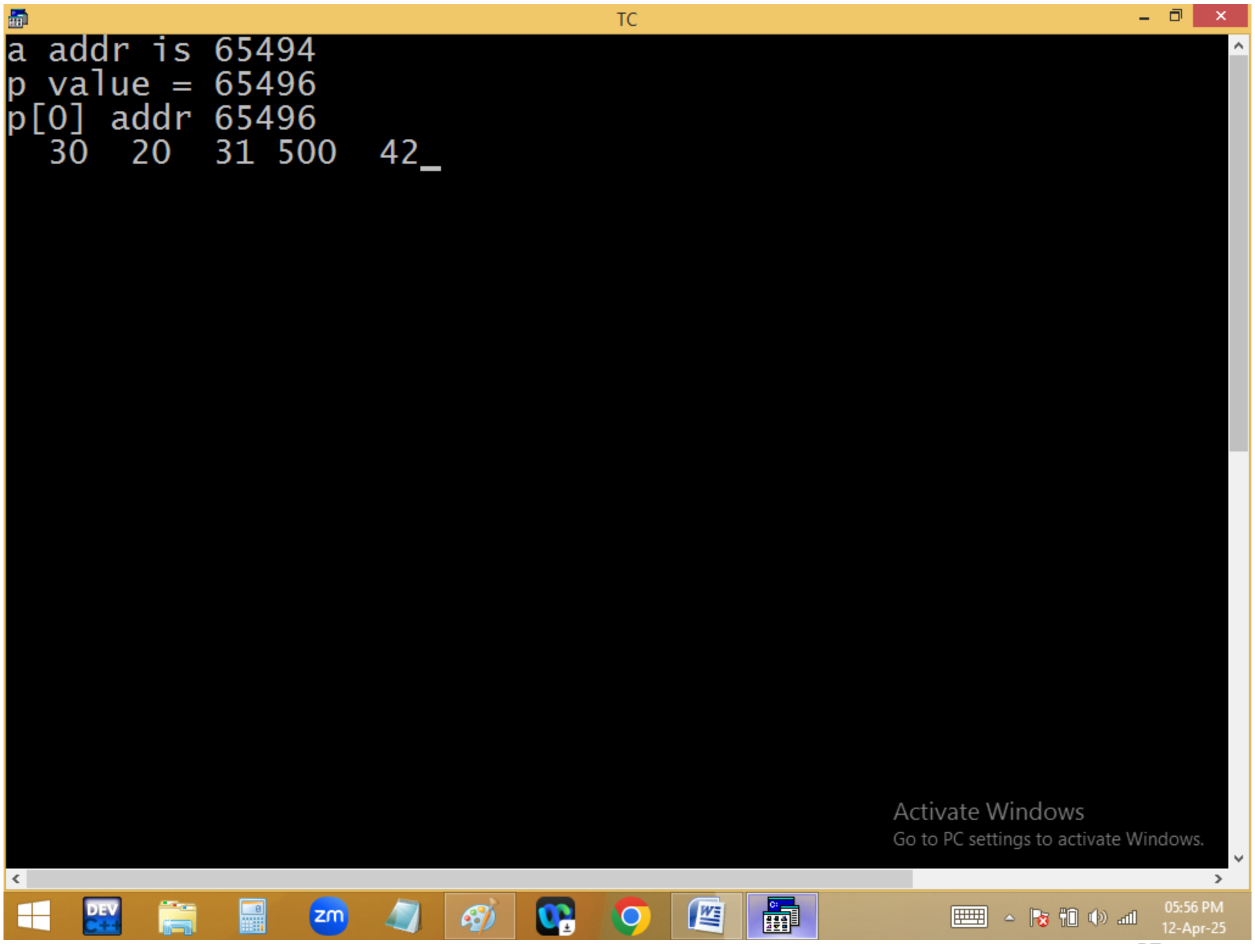
p+=2 ==>65496+2*2=65500

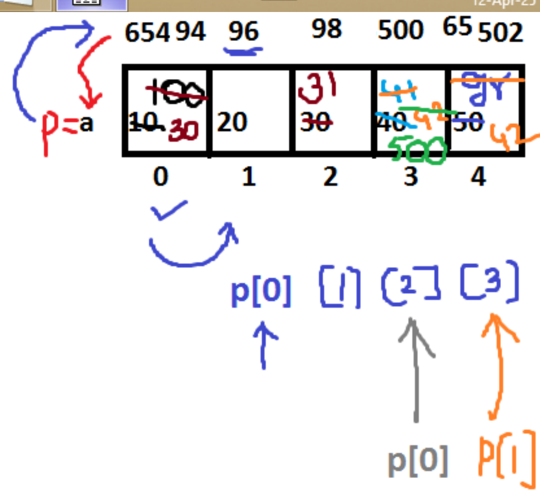
(*p)++ ==> value at 65500++ ==> 40++ ==> 41

***p ==> value at 65500 ==> 41**




```c
#include<stdio.h>
#include<conio.h>
void main()
{
int a[5]={10,20,30,40,50}, *p=&a[0],i;
clrscr();
printf("a addr is %u\n",a);
*p=100;
p[0]=p[2]++;
++p[3];
p[4]=*p++; printf("p value = %u\n",p);
printf("p[0] addr %u\n",&p[0]);
p+=2;
p[1]=++p[0];
*p=500;
for(i=0;i<5;i++)
printf("%4d",a[i]);
getch();
}
```





*p = 100; value at 65494=100

p[0]=p[2]++; p[0]=p[2]==>100=30

==>p[2]++ ==>30++ ==> 31

++p[3] ==> 40++ ==> 41

p[4] = *p++ ==> p[4] = Gr

p+=2 ==>96+2*2=65500

p[1] = ++p[0];

p[1] ==> Gr = ++p[0]==>++41 ==> 42

*p = 500 ==> p[0]=500